

School of Natural Sciences and Mathematics

Actuarial Science (BS)

The Bachelor of Science Actuarial Science (AS) Program at The University of Texas at Dallas is administered through the Department of Mathematical Sciences. Students receive rigorous well-rounded actuarial education with a strong foundation in mathematics, statistics, computer and data science, economics, and finance. All students are prepared to take seven actuarial preliminary Society of Actuaries (SOA) exams: Probability - P/1, Financial Mathematics - FM/2, Fundamentals of Actuarial Mathematics - FAM, Advanced Short Term Actuarial Mathematics - ASTAM, Advanced Long Term Actuarial Mathematics - ALTAM, Statistics for Risk Modeling - SRM, Predictive Analytics - PA. Students achieve Validation of Educational Experience (VEE) credits in Accounting and Finance, Economics, and Mathematical Statistics. Students also receive rigorous instruction in preparation for a major part of the three Casualty Actuarial Society (CAS) Exams: Actuarial Models: Financial Economics - 3F, Modern Actuarial Statistics I and II - MAS I and MAS II. Upon completion of this program, students will have the knowledge and business background necessary to pursue a career as an actuary, as well as to undertake graduate study in actuarial science, statistics, mathematics, economics, data science or finance.

Bachelor of Science in Actuarial Science

[Degree Requirements](#) (120 semester credit hours)

[View an Example of Degree Requirements by Semester](#)

Faculty

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Professors: Swati Biswas, Min Chen, Pankaj Choudhary, Baris Coskunuzer, Mieczyslaw Dabkowski, Vladimir Dragovic, Sam Efromovich, Wieslaw Krawcewicz, Susan Minkoff, L. Felipe Pereira, Dmitry Rachinskiy, Viswanath Ramakrishna, Janos Turi

Associate Professors: Maxim Arnold, Yan Cao, Liang Hong, Oleg Makarenkov, Tomoki Ohsawa, Anh Tran

Assistant Professors: Carlos Arreche, Noirrit Chandra, Ronan Conlon, Rizwanur Khan, Qiwei Li, Stephen McKeown, Chuan-Fa Tang, Jiayi Wang, Nathan Williams, Nan Wu

Professors Emeriti: Larry Ammann, Ali Hooshyar, Patrick Odell, John Van Ness

Clinical Professor: Natalia Humphreys

Clinical Associate Professor: Mohammad Akbar

Clinical Assistant Professor: Wenyi Lu

Professors of Instruction: Anatoly Eydelzon, Manjula Foley, Bentley Garrett, Yuly Koshevnik

Associate Professors of Instruction: Mohammad Ahsan, Kelly Aman, Malgorzata Dabkowski, Rabin Dahal, Derege Mussa, My Linh Nguyen, Jigarkumar Patel, Julie Sutton

Assistant Professors of Instruction: Anani Komla Adabrah, Iris Alvarado, Saikat Biswas, Hui Ding, Kemelli Estacio-Hiroms, Huizhen Guo, Shengjie Jiang, Joselle Kehoe, Runzhou Liu, Neha Makhijani, Irina Martynova, Diarisoa Mihaja Rakotomalala, Adrian Murza, Ajaya Paudel, Octavious Smiley, Nasrin Sultana, Che-Yu Wu

I. Core Curriculum Requirements: 42 semester credit hours

Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

Communication: 6 semester credit hours

Select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

Mathematics: 3 semester credit hours

[MATH 2417](#) Calculus I ^{1 2 3}_{,-,-}

Or select any 3 semester credit hours from [Mathematics Core](#) courses (see advisor)

Life and Physical Sciences: 6 semester credit hours ^{1 4}_{,-,-}

[PHYS 2325](#) Mechanics

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat ^{5 6}_{,-,-}

or [CHEM 1311](#) General Chemistry I

or [CHEM 1315](#) Honors Freshman Chemistry I

[PHYS 2326](#) Electromagnetism and Waves

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves ⁵_{,-,-}

or [CHEM 1312](#) General Chemistry II

or [CHEM 1316](#) Honors Freshman Chemistry II

Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor)

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

American History: 6 semester credit hours

Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

Government/Political Science: 6 semester credit hours

Select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

[ECON 2302](#) Principles of Microeconomics¹_{_,_,_}

Or select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

Component Area Option: 6 semester credit hours

[MATH 2417](#) Calculus I^{1 2 3}_{_,_,_}

[MATH 2419](#) Calculus II^{1 2 3}_{_,_,_}

[PHYS 2125](#) Physics Laboratory I^{1 4}_{_,_}

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat^{5 6}_{_,_}

or [CHEM 1111](#) General Chemistry Laboratory I^{1 4}_{_,_}

Or select any 6 semester credit hours from [Component Area Option](#) courses (see advisor)

II. Major Requirements: 77-78 semester credit hours

Major Preparatory Courses: 29-30 semester credit hours beyond Core Curriculum

[NATS 1101](#) Natural Sciences and Mathematics Freshman Seminar

[ACCT 2301](#) Introductory Financial Accounting

[ACCT 2302](#) Introductory Management Accounting

[BCOM 3300](#) Business Communication

[CS 1436](#) Programming Fundamentals

[CS 1337](#) Computer Science I

or [MATH 2370](#) Introduction to Programming with MATLAB⁷

[ECON 2301](#) Principles of Macroeconomics

[ECON 2302](#) Principles of Microeconomics

[MATH 2417](#) Calculus I^{1 2 3}

[MATH 2418](#) Linear Algebra

[MATH 2419](#) Calculus II^{1 2 3}

[MATH 2420](#) Differential Equations with Applications

[PHYS 2325](#) Mechanics^{1 4}

and [PHYS 2125](#) Physics Laboratory I^{1 4}

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat^{1 5 6 8}

or [CHEM 1311](#) General Chemistry I^{1 4}

and [CHEM 1111](#) General Chemistry Laboratory I^{1 4}

or [CHEM 1315](#) Honors Freshman Chemistry I^{1 4}

and [CHEM 1115](#) Honors Freshman Chemistry Laboratory I

[PHYS 2326](#) Electromagnetism and Waves^{1 4}

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves^{1 4 5}

or [CHEM 1312](#) General Chemistry II^{1 4}

or [CHEM 1316](#) Honors Freshman Chemistry II^{1 4}

[PHYS 2126](#) Physics Laboratory II

or [CHEM 1112](#) General Chemistry Laboratory II

or [CHEM 1116](#) Honors Freshman Chemistry Laboratory II

Major Core Courses: 48 semester credit hours

[ACTS 4301](#) Long Term Actuarial Mathematics I

[ACTS 4302](#) Investment and Financial Markets

[ACTS 4303](#) Long Term Actuarial Mathematics II

[ACTS 4304](#) Short Term Actuarial Mathematics I

[ACTS 4305](#) Short Term Actuarial Mathematics II

[ACTS 4307](#) Statistics for Risk Modeling

[ACTS 4308](#) Actuarial Financial Mathematics

[ACTS 4310](#) Predictive Analytics

[ACTS 4311](#) Theory of Credibility

[MATH 3351](#) Advanced Calculus

[ITSS 3300](#) Information Technology for Business

[ITSS 4301](#) Database Systems

[STAT 3355](#) Introduction to Data Analysis

[STAT 4351](#) Probability

[STAT 4352](#) Mathematical Statistics

[STAT 4382](#) Stochastic Processes

III. Elective Requirements: 0-1 semester credit hour

[RMIS 3370](#) Principles of Risk Management and Insurance

[RMIS 4331](#) Business Liability Risk Management and Insurance

[MATH 3310](#) Theoretical Concepts of Calculus

[MATH 3311](#) Abstract Algebra I

[MATH 3379](#) Complex Variables

[MATH 4334](#) Numerical Analysis

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

Preparation for Actuarial Exams

Exam 1/P: [STAT 4351](#)

Exam 2/FM: [ACTS 4308](#)

Exams 3L/FAM/ALTAM: [ACTS 4301](#) and [ACTS 4303](#)

Exams 4/FAM/ASTAM: [ACTS 4304](#) and [ACTS 4305](#) and [ACTS 4311](#)

Exam SRM: [ACTS 4307](#)

Exam PA: [ACTS 4310](#)

Validation by Educational Experience (VEE) Credits

Mathematical Statistics: [STAT 3355](#) and [STAT 4352](#)

Accounting and Corporate Finance: [ACCT 2302](#) and [ACTS 4302](#)

Economics: [ECON 2301](#) and [ECON 2302](#)

Fast Track Baccalaureate/Master's Degrees

In response to the need for post-baccalaureate education, a Fast Track program is available to well-qualified UT Dallas undergraduate students. Qualified seniors may take up to 15 graduate semester credit hours that may be used to complete the baccalaureate degree and also to satisfy the requirements for the master's degree. Interested students should see the Associate Dean of Undergraduate Education (ADU) for specific requirements.

1. A required Major preparatory course that also fulfills a Core Curriculum requirement. Semester credit hours are counted in Core Curriculum.
2. Three semester credit hours of Calculus are counted to fulfill the Mathematics Core Requirement with the remaining one semester credit hour to be counted under Component Area Option Core Requirement.

3. Students may choose one of the following calculus sequences: (a) MATH 2413, MATH 2414, and MATH 2415; or (b) MATH 2417 and MATH 2419.
4. Six semester credit hours of Physics or Chemistry are counted under Science core, and one semester credit hour of Physics or Chemistry (PHYS 2125 or CHEM 1111) are counted under Component Area Option core.
5. Please consult your advisor if selecting Honors Physics.
6. Students may use three semester credit hours of PHYS 2421 to count under Science core, and one semester credit hour of PHYS 2421 under Component Area Option core.
7. MATH 2370 will provide basic knowledge of a programming and numeric computing platform MATLAB and a better preparation for MATH 4334 if elected.
8. Students who complete PHYS 2421 do not need to complete PHYS 2125.

Minors

Students must take a minimum of 18 semester credit hours for the minor, 12 of which must be upper-division semester credit hours. Students who take a minor will be expected to meet the normal prerequisites in courses making up the minor, and should maintain a minimum GPA of 2.000 on a 4.00 scale (C average). Semester credit hours may not be used to satisfy both the major and minor requirements; however, free elective semester credit hours or major preparatory classes may be used to satisfy the minor.

For all minors in the School of Natural Sciences and Mathematics students must complete all prerequisite sequences for required minor courses.

Minor in Actuarial Science

23 semester credit hours

The Minor in Actuarial Science program at UT Dallas is administered through the Department of Mathematical Sciences. It is ideal for students who are interested in broadening their experience and knowledge base in the study and analysis of principles of Actuarial Science. The minor core courses prepare students for a number of actuarial exams required for a designation of Associate of the Society of Actuaries, Casualty Actuarial Society, or Canadian Institute of Actuaries. All of the courses in the minor serve as starting points for learning the concepts covered on the preliminary actuarial exams (P/1 - Probability, FM/2 - Financial Mathematics, portions of FAM-L - Fundamentals of Actuarial Mathematics, ALTAM - Advanced Long Term Actuarial Mathematics) and Finance VEE (Validation by Educational Experience) requirement.

Students not majoring in Actuarial Science may obtain a minor in Actuarial Science by satisfying 23

semester credit hours (11 semester credit hours of minor core courses and 12 semester credit hours of minor preparatory courses).

Minor Preparatory Courses (11 semester credit hours)⁹

[MATH 2415](#) Calculus of Several Variables

or [MATH 2419](#) Calculus II

[MATH 3351](#) Advanced Calculus

[MATH 2418](#) Linear Algebra

Minor Core Courses (12 semester credit hours)¹⁰

[STAT 4351](#) Probability

[ACTS 4301](#) Long Term Actuarial Mathematics I

[ACTS 4302](#) Investment and Financial Markets

[ACTS 4308](#) Actuarial Financial Mathematics

9. Students whose major does not require MATH 2417 and MATH 2419 as part of their Mathematics and Component Area Option Core Curriculum Requirements, should take this sequence as their core curriculum courses to ensure efficiency toward the minor.

10. These classes prepare for the major parts of three preliminary actuarial examinations administered by the Society of Actuaries (SOA), Casualty Actuarial Society (CAS), and the Canadian Institute of Actuaries (CIA).

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